



**HKU  
Med**

School of Clinical Medicine  
Department of Medicine  
香港大學內科學系

# Urticaria, angioedema and anaphylaxis

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# Learning objectives

- Define and describe the basic features of urticaria, wheals, angioedema and anaphylaxis.
- Able to differentiate between histamine- and bradykinin-mediated angioedema.
- List the common causes and investigations of urticaria.
- Appreciate the epidemiology and common causes of anaphylaxis in Hong Kong.
- Understand pathophysiology and possible causes of bradykinin-mediated angioedema.
- Describe treatment approach to chronic spontaneous urticaria.
- Understand the approach to management of histaminergic urticaria and anaphylaxis.

# Definitions - *not everything is “allergy”*

- **Wheals** (hives)
  - (1) Central swelling (2) pruritus (3) a *fleeting* nature (<24 hours)
    - *c.f urticarial vasculitis = painful, >24 hours, constitutional symptoms, post-inflammatory hyperpigmentation*
- **Angioedema** = swelling of the deep dermis/subcutaneous tissue, Painful rather than pruritic, slower resolution ~up to 72 hours
  - Important to differentiate: with or without wheals
- **Anaphylaxis** = serious, acute and life-threatening systemic hypersensitivity reaction
  - Various definitions
  - Usually involvement of two organ systems or more
  - Especially with cardiac or respiratory involvement



# Anaphylaxis

## World Allergy Organization Anaphylaxis Guidance 2020

**Anaphylaxis is highly likely when any one of the following 2 criteria are fulfilled:**

1. Acute onset of an illness (minutes to several hours) with simultaneous involvement of the skin, mucosal tissue, or both (eg, generalized hives, pruritus or flushing, swollen lips-tongue-uvula)

**AND AT LEAST ONE OF THE FOLLOWING:**

a. Respiratory compromise (eg, dyspnea, wheeze-bronchospasm, stridor, reduced PEF, hypoxemia)

b. Reduced BP or associated symptoms of end-organ dysfunction (eg, hypotonia [collapse], syncope, incontinence)

c. Severe gastrointestinal symptoms (eg, severe crampy abdominal pain, repetitive vomiting), especially after exposure to non-food allergens

2. Acute onset of **hypotension<sup>a</sup> or bronchospasm<sup>b</sup> or laryngeal involvement<sup>c</sup> after exposure<sup>d</sup>** to a known or highly probable allergen<sup>d</sup> for that patient (minutes to several hours), even in the absence of typical skin involvement.

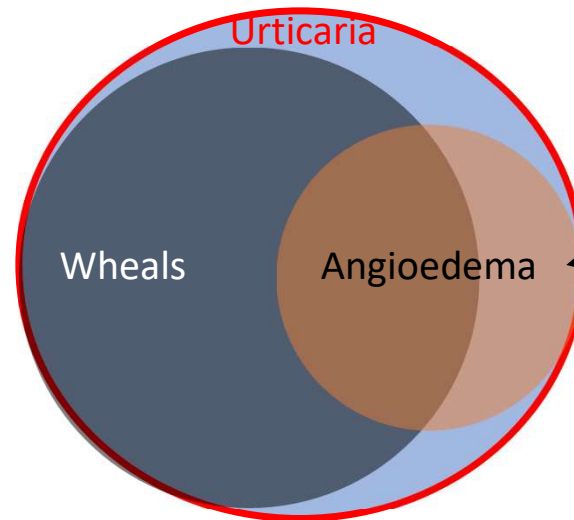
**Table 2.** Amended criteria for the diagnosis of anaphylaxis. PEF, Peak expiratory flow; BP, blood pressure. a. Hypotension defined as a decrease in systolic BP greater than 30% from that person's baseline, OR i. Infants and children under 10 years: systolic BP less than  $(70 \text{ mmHg} + [2 \times \text{age in years}])$  ii. Adults and children over 10 years: systolic BP less than  $<90 \text{ mmHg}$ . b. Excluding lower respiratory symptoms triggered by common inhalant allergens or food allergens perceived to cause "inhalational" reactions in the absence of ingestion. c. Laryngeal symptoms include: stridor, vocal changes, odynophagia. d. An allergen is a substance (usually a protein) capable of triggering an immune response that can result in an allergic reaction. Most allergens act through an IgE-mediated pathway, but some non-allergen triggers can act independent of IgE (for example, via direct activation of mast cells). Adapted from <sup>(26)</sup>

# Since 2018, urticaria = wheals and/or angioedema

## Definition

Urticaria is a condition characterized by the development of wheals (hives), angioedema or both.

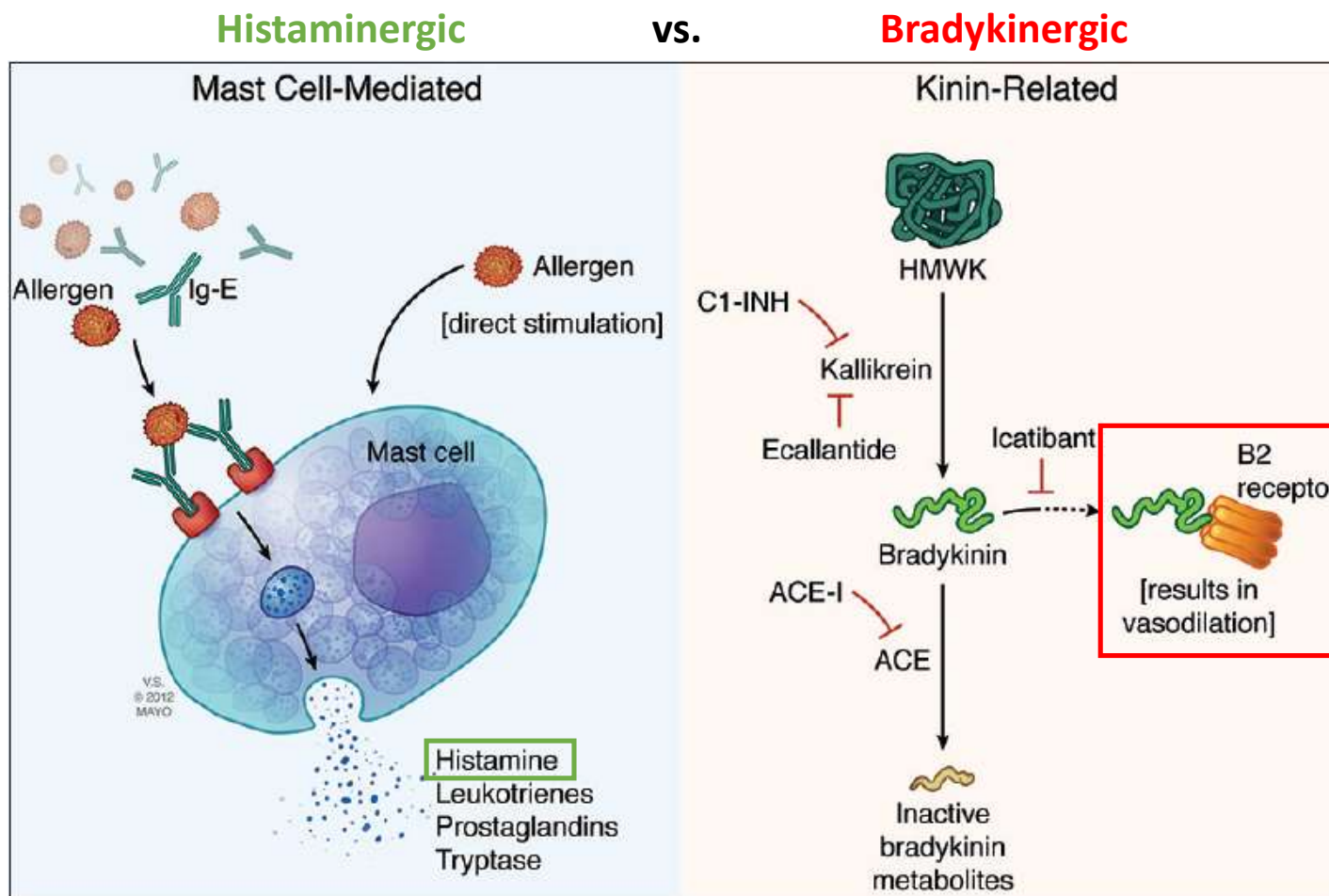
Urticaria is not simply an eruption of wheals!



**Angioedema with wheals**  
**vs.**  
**Angioedema without wheals**

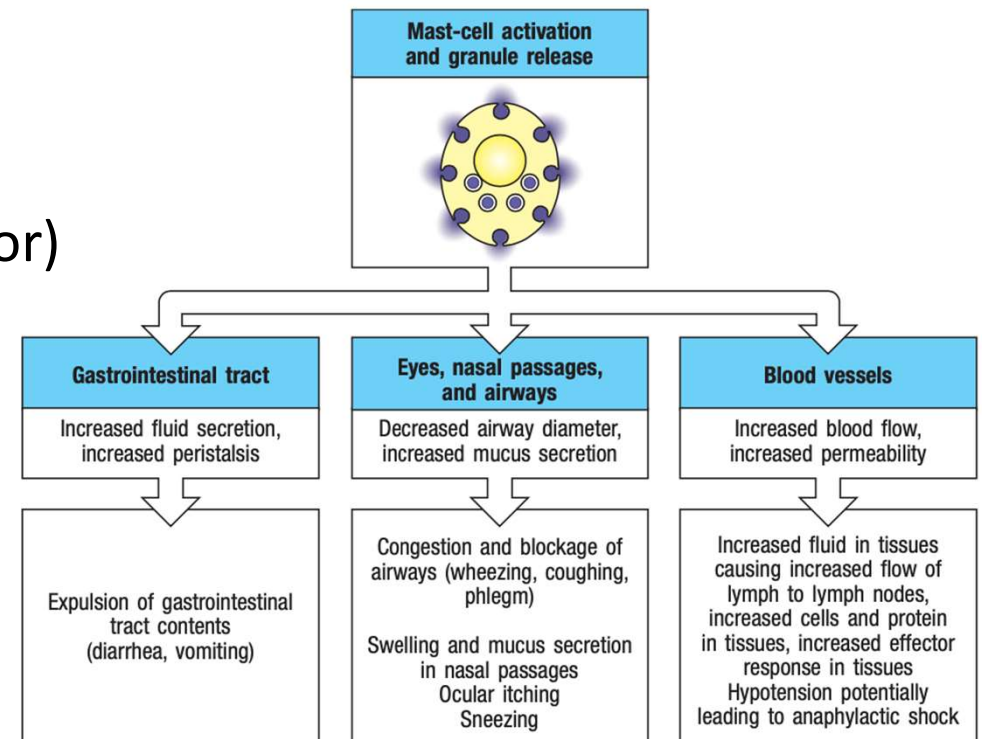
*Pattern is important to distinguish between **different pathomechanisms** → different clinical course and management*

# Pathomechanisms (Endotypes)



# Histaminergic pathway

- **Mast cell** mediated
- Activated via  $Fc\epsilon R1\alpha$
- Release of inflammatory mediator  
e.g. histamine, PAF (platelet activating factor)
  - Induce sensory nerve stimulation (pruritis)
  - Increased vascular permeability
  - Vasodilation
  - Contraction of bronchial & intestinal smooth muscle
- Can be “allergic” / “non-allergic”

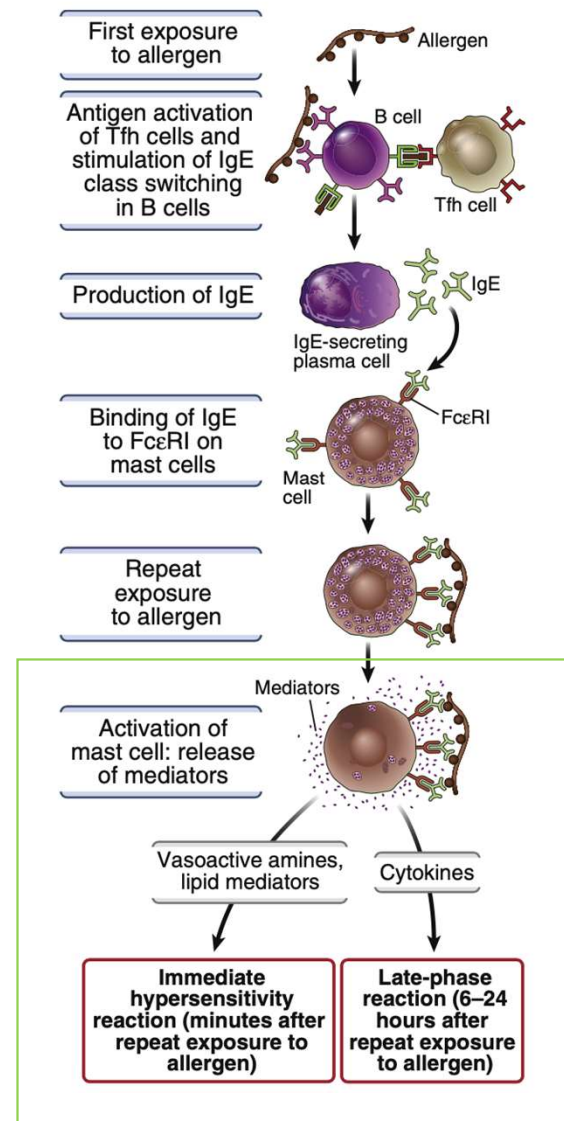




# Histaminergic

## “Allergic” reactions

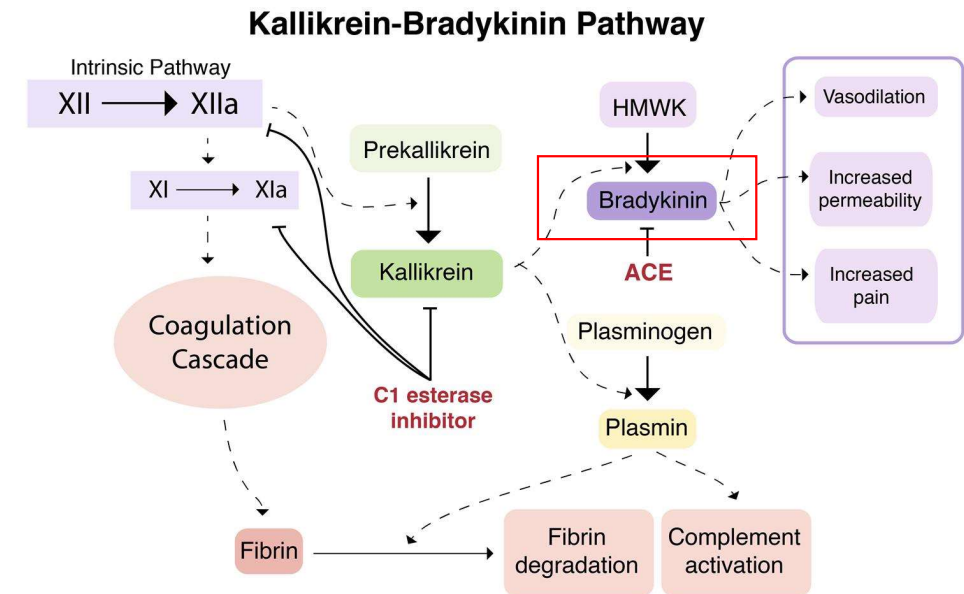
- IgE mediated = “allergic”
- Type I hypersensitivity
- Binding of the allergen cross-links IgE antibodies on mast cells
- Triggers FcεRI
- Implicated in: food allergies, drug allergies, allergic rhinitis, asthma and **anaphylaxis**





# Bradykinergic pathway

- Mast cell **independent**
- Bradykinin generated through activation of plasma contact system
- A vasoactive molecule:
  - Vasodilation
  - Swelling
  - Pain
- Regulated by **C1-esterase inhibitor**



# Bradykinergic angioedema

- Characterized by swelling attacks (angioedema)
- **Absence of wheals** and pruritis
- May involve face, extremities, abdomen, and **most dangerously: upper airway**
- Can be **drug-related (ACEi)**, hereditary, or acquired

**Beware of HAE**  
(Hereditary Angioedema)



## WHAT A HEREDITARY ANGIOEDEMA (HAE) ATTACK LOOKS LIKE

Facial attacks



Peripheral attacks

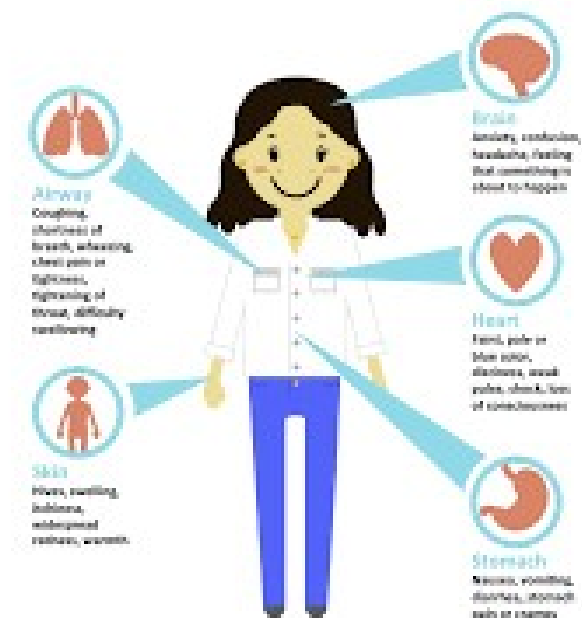
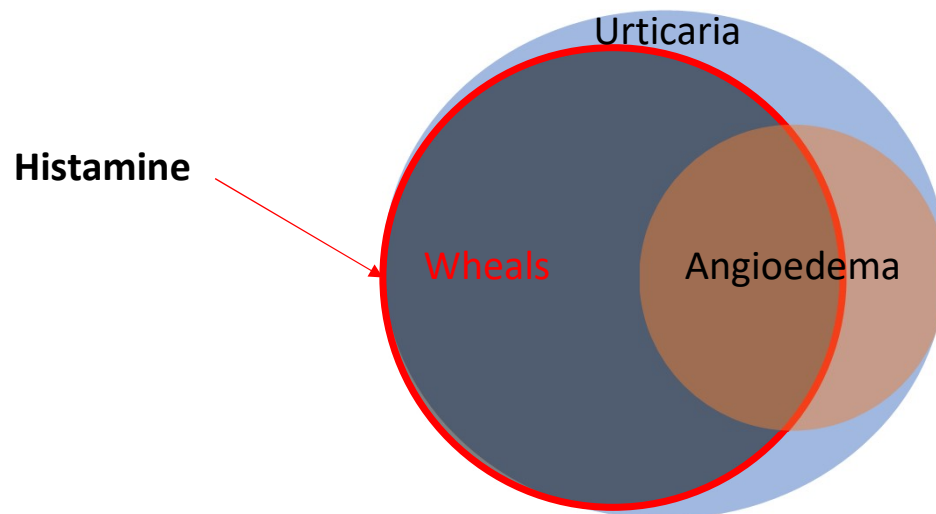


Abdominal attacks



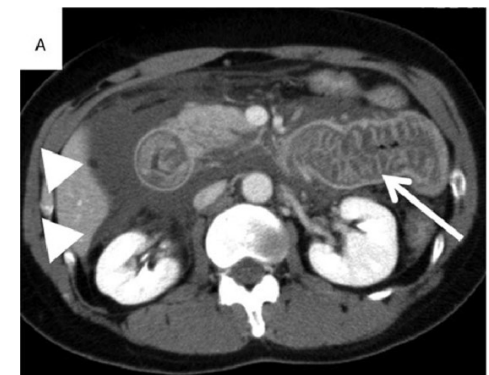
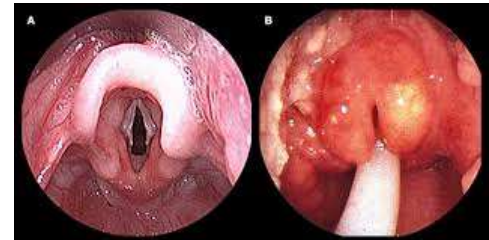
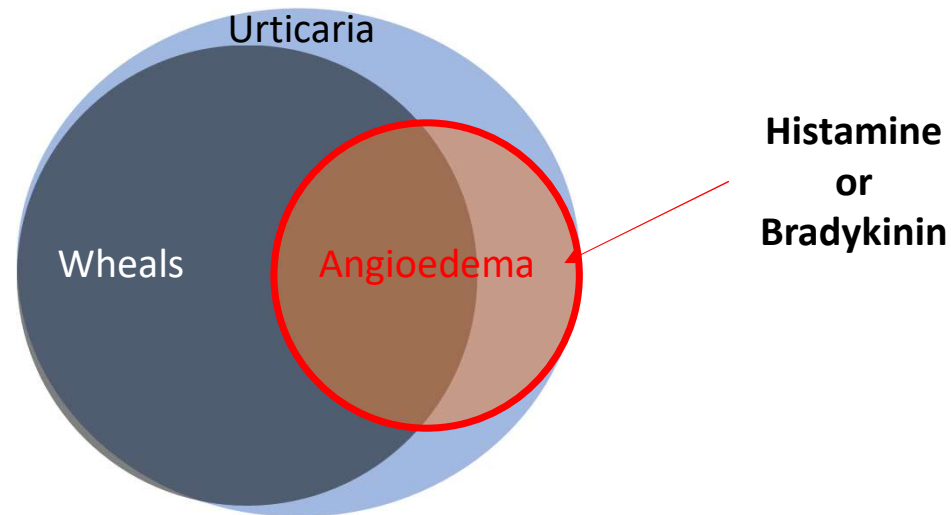
# Clinical classification (Phenotypes)

Wheals  $\pm$  *angioedema*  
Histaminergic

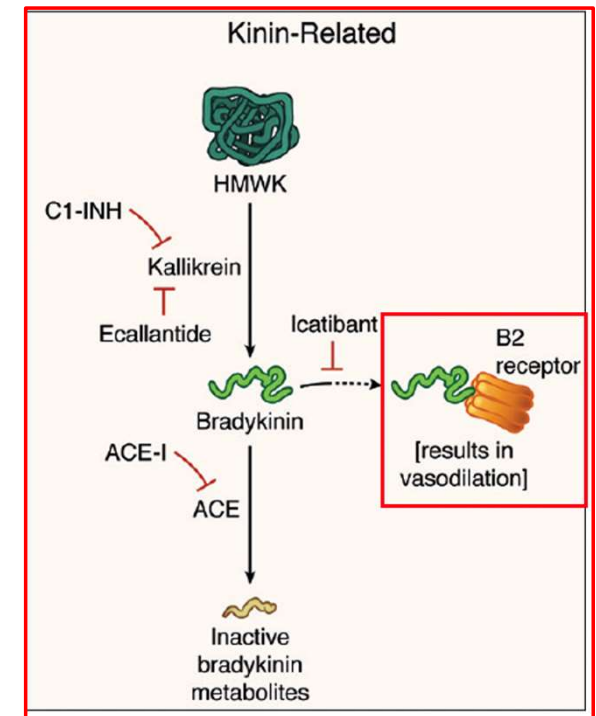
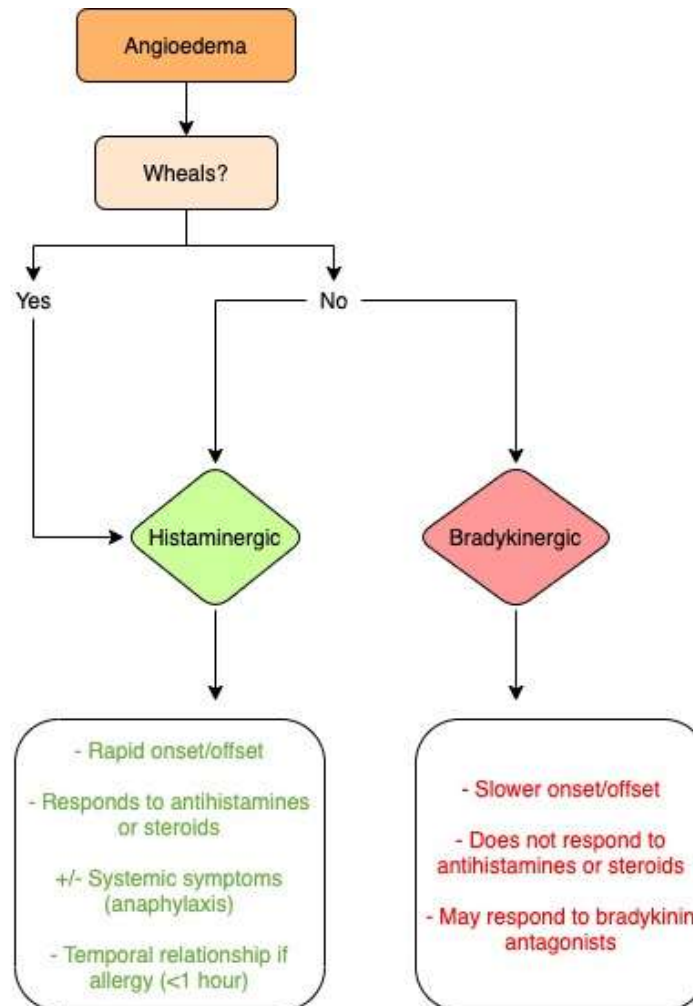
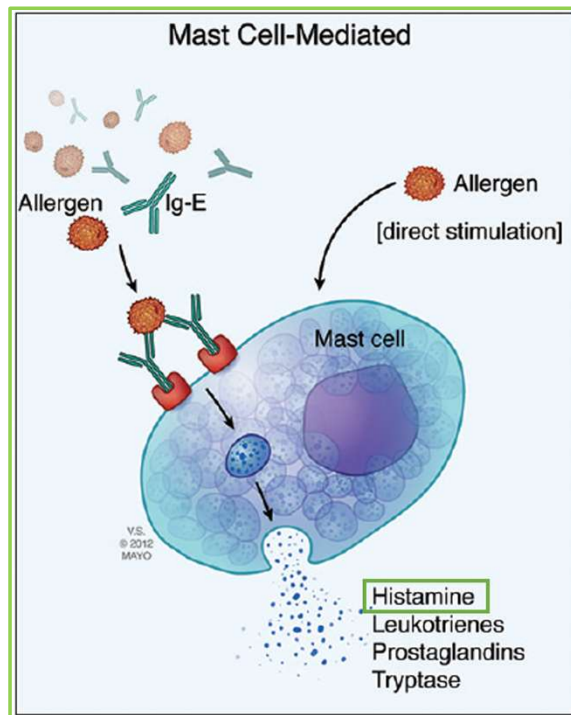


# Clinical classification (Phenotypes)

Angioedema without wheals  
Histaminergic or Bradykinergic



# Classifying angioedema by endotype (mechanism)





## How to clinically differentiate histaminergic and bradykinergic angioedema?

Clinical differences between histaminergic and bradykinergic angioedema can be illustrated in the table below.<sup>3,4</sup>

| Features                                                | Histaminergic                                  | Bradykinergic                                                                                                                    |
|---------------------------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Concomitant hives?                                      | Yes, although not always                       | No                                                                                                                               |
| Any gastrointestinal involvement?                       | Uncommon, possible in cases of anaphylaxis     | Yes, bowel edema can commonly present with gastrointestinal upset. In severe cases, can lead to bowel obstruction and ischaemia. |
| Any respiratory involvement?                            | Uncommon, but possible in cases of anaphylaxis | Possible, due to laryngeal oedema                                                                                                |
| Duration of symptoms                                    | Shorter, usually subside within 1-3 days       | Longer, can last for 3-4 days                                                                                                    |
| Response to antihistamines, steroids or adrenaline      | Yes                                            | No                                                                                                                               |
| Response to bradykinin inhibitors or C1-INH replacement | No                                             | Yes                                                                                                                              |

Did you know that ...

**NOT ALL SWELLING IS AN ALLERGIC REACTION**



HAE presents with

- swelling
- abdominal pain
- airway obstruction

→ often misdiagnosed as an allergic reaction

Find out more: [haei.org](http://haei.org)

**HAEi**  
HAE International

**Approach to Angioedema in Hong Kong**

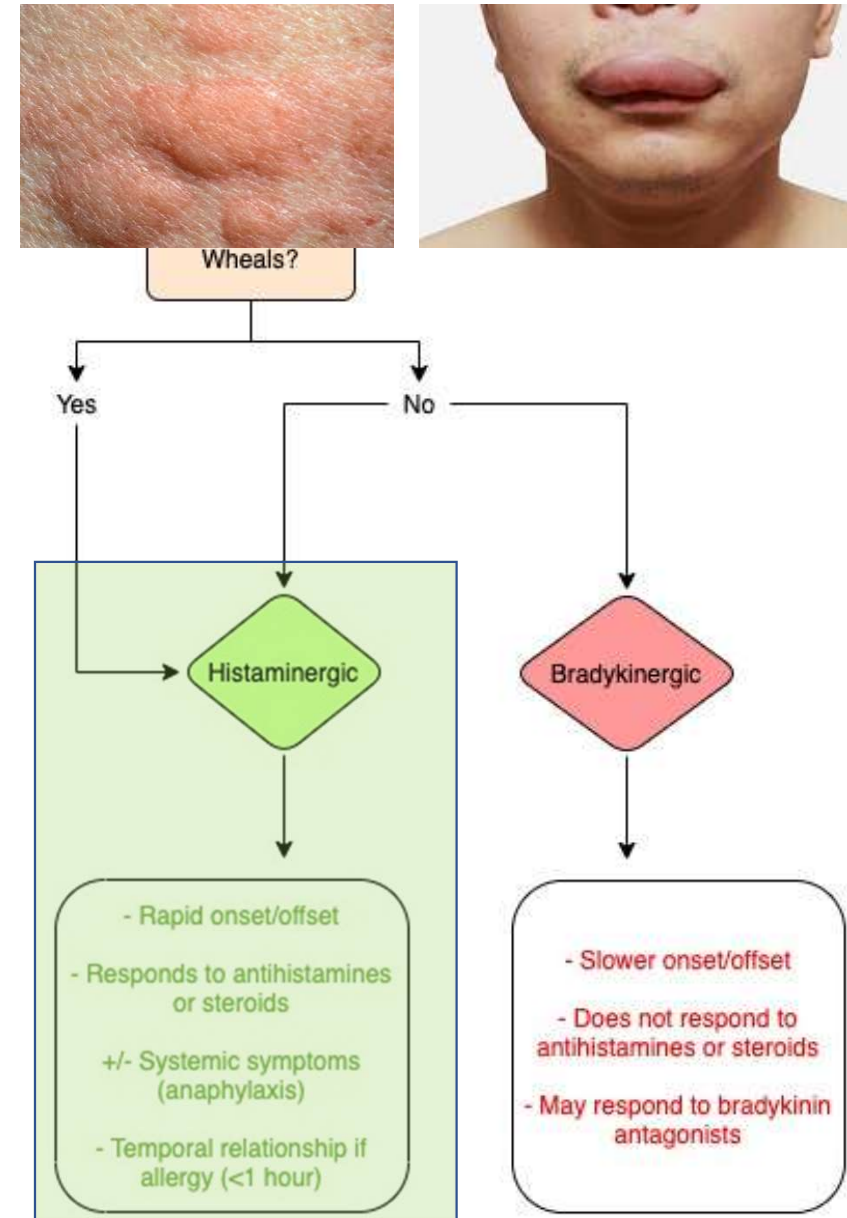


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香港大學內科學系

<http://imm.hku.hk>

# Differentials: histaminergic

- “Allergic” type I hypersensitivity reaction; e.g. foods, drugs, venom
  - Mostly IgE-mediated reactions
- Spontaneous or autoimmune urticaria and angioedema (CSU)
  - Anti-FcεR1α antibody (mast cells)
- Inducible urticaria and angioedema
- Urticarial vasculitis
- Auto-inflammatory syndromes





# Histaminergic Anaphylaxis

Differentials: histaminergic

“Allergic” type I hypersensitivity reaction; e.g. foods, drugs, venom

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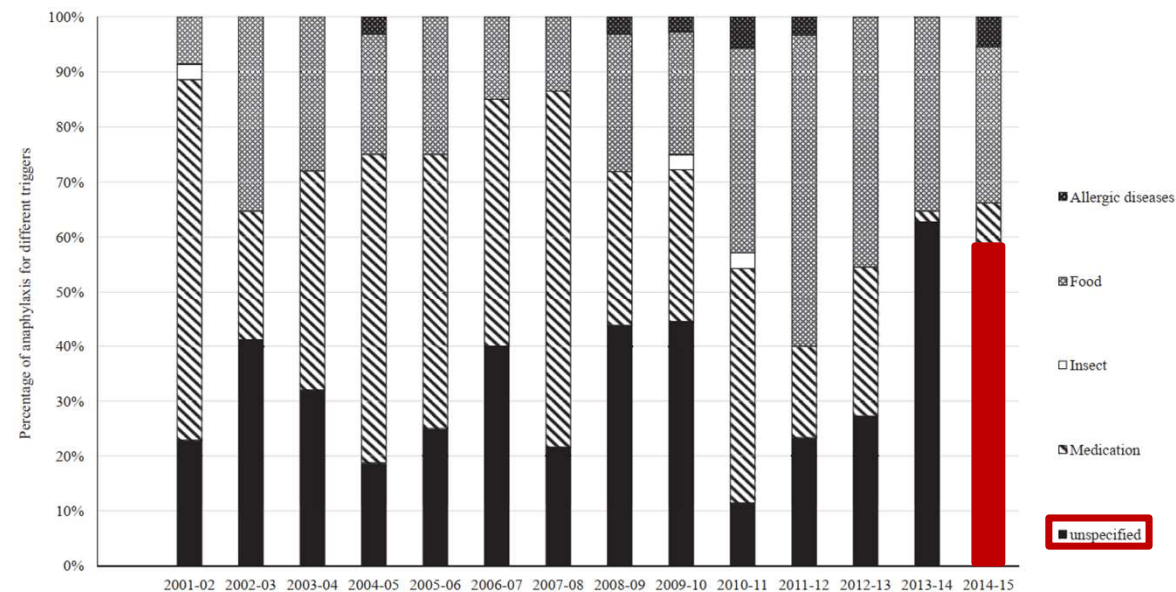
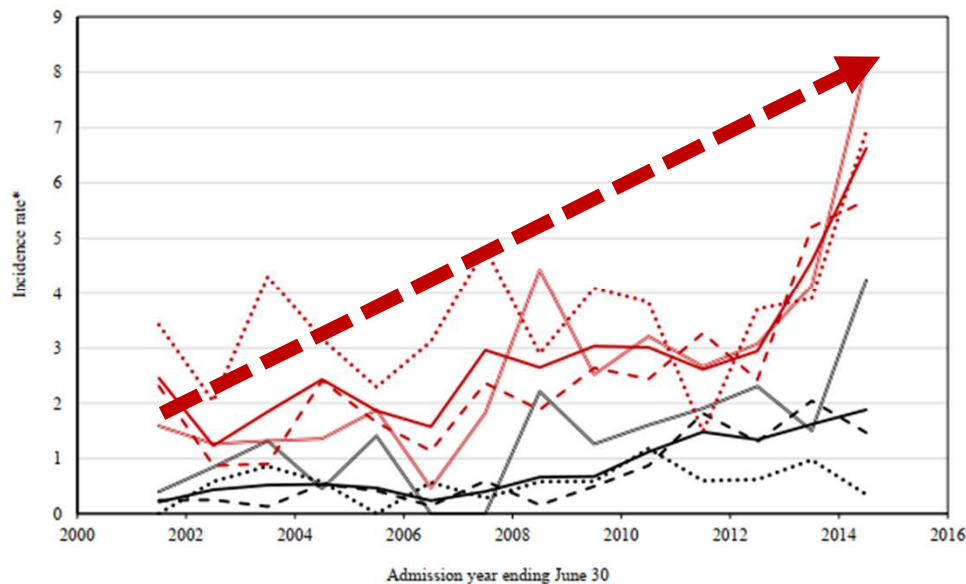
*The Journal of Allergy and Clinical Immunology:*

## In Practice

### Increasing hospital presentations for anaphylaxis in the pediatric population in Hong Kong

Yichao Wang, MSc • Jennifer Julia Koplin, BSc (Hons), PhD • Marco Hok Kung Ho, MD, FRCPCH •

Wilfred Hing Sang Wong, MMedSc • Katrina Jane Allen, BMedSc, MBBS, FRACP, PhD



# Histaminergic

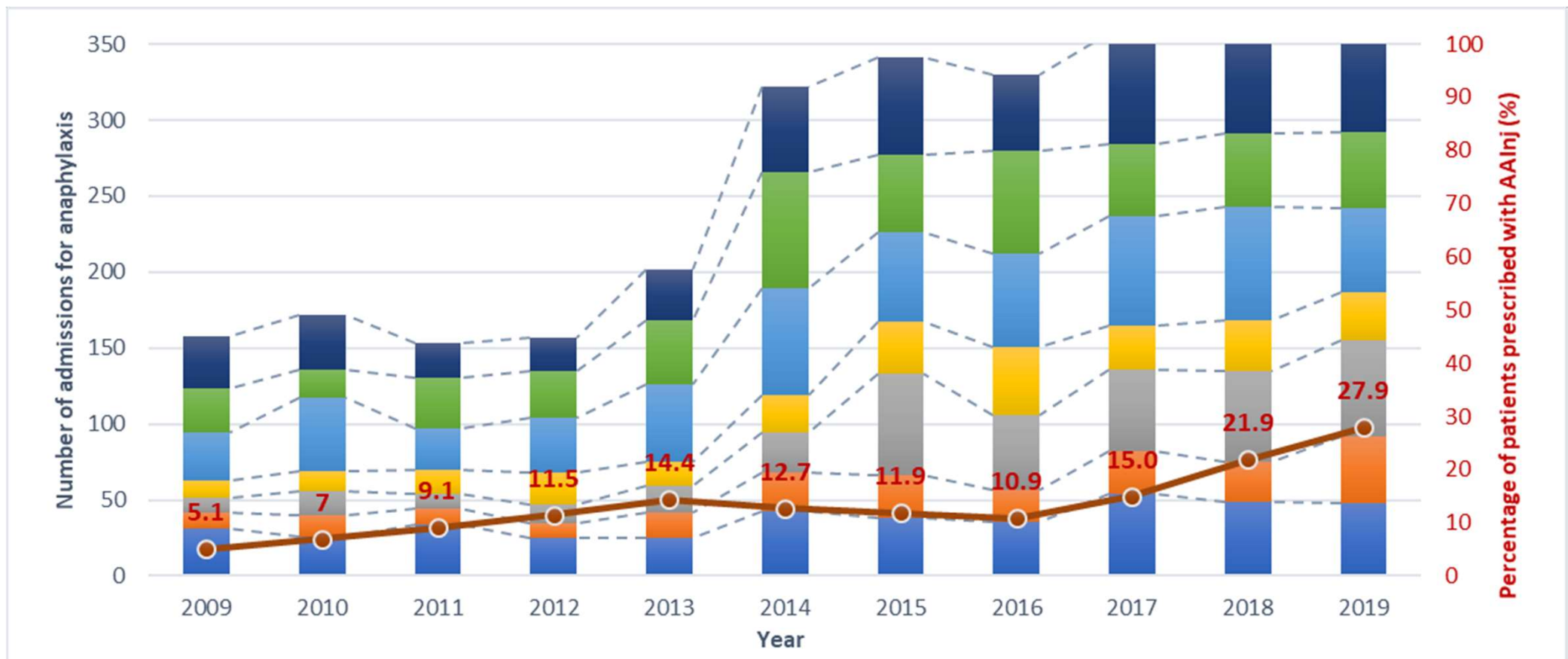
## Anaphylaxis in Hong Kong

### RESEARCH

### Open Access

Increasing incidence of anaphylaxis in Hong Kong from 2009 to 2019—discrepancies of anaphylaxis care between adult and paediatric patients

Philip Hei Li<sup>1†\*</sup>, Agnes S. Y. Leung<sup>2†</sup>, Rebecca M. Y. Li<sup>2</sup>, Ting-fan Leung<sup>2</sup>, Chak-sing Lau<sup>1</sup> and Gary W. K. Wong<sup>2</sup>



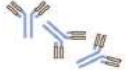
# Histaminergic

## Anaphylaxis - immunology

IgE mediated

Non-IgE mediated

~20% idiopathic

| IMMUNOLOGIC MECHANISMS (IgE dependent)                                                                                                            |                                                                                                               |                                                                                                                                       |                                                                                                                                              |                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <br>peanut                                                     | <br>tree nuts              | <br>shellfish                                      | <br>fish                                                  | <br>stinging insects     |
| <br>milk                                                       | <br>egg                    | <br>soybean                                        | <br>peach                                                 | <br>sesame               |
| Foods                                                                                                                                             |                                                                                                               |                                                                                                                                       | Venoms                                                                                                                                       | Medications*                                                                                                |
| <br>Natural rubber latex                                       | <br>Occupational allergens | <br>Seminal fluid                                  | <br>Aeroallergens                                         | <br>Radiocontrast media* |
| IMMUNOLOGIC MECHANISMS (IgE independent)                                                                                                          |                                                                                                               |                                                                                                                                       |                                                                                                                                              |                                                                                                             |
| <br>Radiocontrast media*                                       | <br>NSAIDs* **             | <br>Dextrans<br>(e.g. HMW*** iron or other source) | <br>Biologic agents*<br>(e.g. some monoclonal antibodies) |                                                                                                             |
| NONIMMUNOLOGIC MECHANISMS (Direct mast cell activation)                                                                                           |                                                                                                               |                                                                                                                                       |                                                                                                                                              |                                                                                                             |
| <br>Physical factors<br>(e.g. exercise, cold, heat, sunlight) | <br>Ethanol               |                                                                                                                                       | <br>Medications*<br>(e.g. opioids)                       |                                                                                                             |
| IDIOPATHIC ANAPHYLAXIS (No apparent trigger)                                                                                                      |                                                                                                               |                                                                                                                                       |                                                                                                                                              |                                                                                                             |
| <br>Previously unrecognized allergen?                        |                                                                                                               | <br>Mastocytosis/clonal mast cell disorder?      |                                                                                                                                              |                                                                                                             |
| *Trigger anaphylaxis by more than one mechanism    **NSAIDs, non-steroidal anti-inflammatory drugs    ***HMW, high molecular weight               |                                                                                                               |                                                                                                                                       |                                                                                                                                              |                                                                                                             |

Histaminergic

*A diagnosis of exclusion!*

**Idiopathic ...?**



**Keep on looking for  
trigger(s)/explanation!**

- **Hidden allergens vs. Anaphylaxis mimics:**  
acute urticaria/angioedema, asthmatic attacks, vasovagal syncope, panic attacks, shock and other causes of sudden collapse/respiratory distress.....

# Histaminergic

## Anaphylaxis in Hong Kong

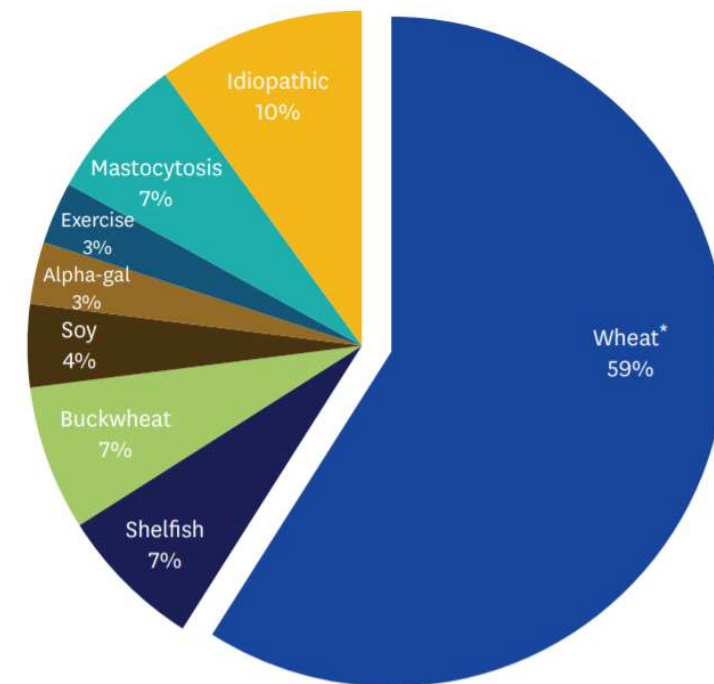
- HKU study published in 2020
- From 2018-19: **>10% of adult patients** referred to Queen Mary Hospital Immunology Clinic were for “**idiopathic**” anaphylaxis (i.e. unknown cause)
- After seen by allergist with complete workup, **specific causes could be identified in 90%**
- Despite surviving anaphylaxis, less than 1/3 of patients were prescribed AAI prior to consultation
- Compared to patients in UK, HK patients had more urticaria and cardiovascular manifestations

### Differences in omega-5-gliadin allergy: East versus West

Philip Hei Li <sup>1,\*</sup>, Iason Thomas <sup>2,†</sup>, Jane Chi-Yan Wong <sup>1</sup>, Krzysztof Rutkowski <sup>2</sup>,  
and Chak-Sing Lau <sup>1</sup>

<sup>1</sup>Division of Rheumatology and Clinical Immunology, Department of Medicine, Queen Mary Hospital, University of Hong Kong, Hong Kong

<sup>2</sup>Department of Allergy, Guy's and St Thomas' NHS Foundation Trust, London, United Kingdom



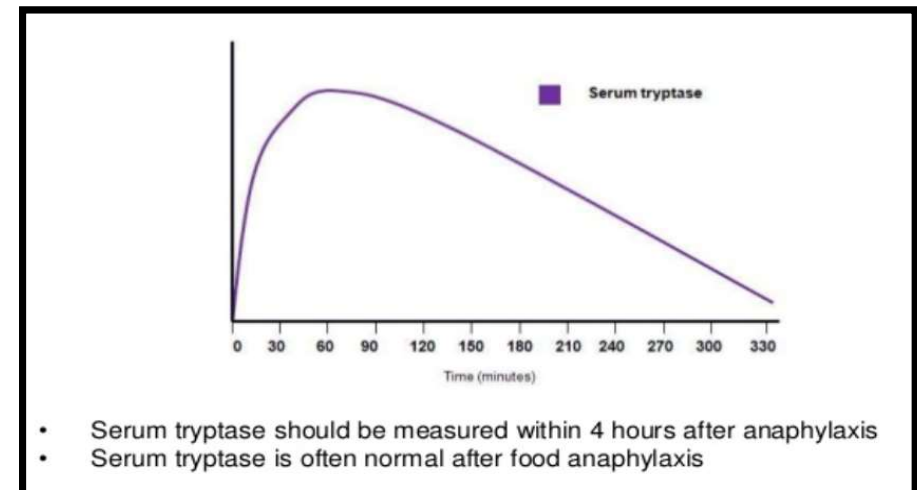
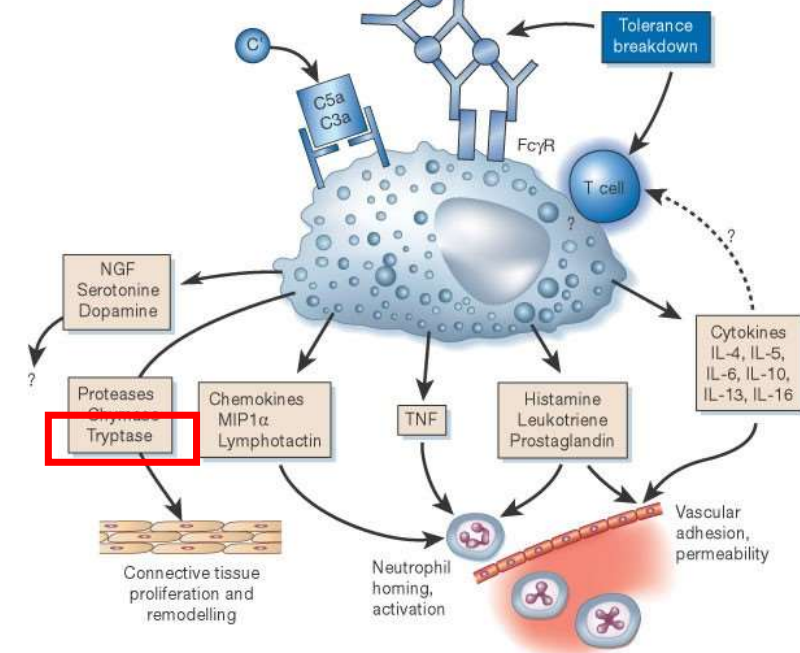


# Histaminergic

## Anaphylaxis: management

- **Acute serum tryptase**  
(within 6 hours of onset)
- Ascertain any potential allergens  
(**within one hour of onset**)
  - Foods, drugs, venoms
  - Co-factors: exercise, NSAIDs, alcohol, menstruation, illness, stress
- Treat with **adrenaline as first line**  
(possible ADJUNCTS: anti-histamines / steroids)

Benoist, C., Mathis, D. Mast cells in autoimmune disease. *Nature* (2002)

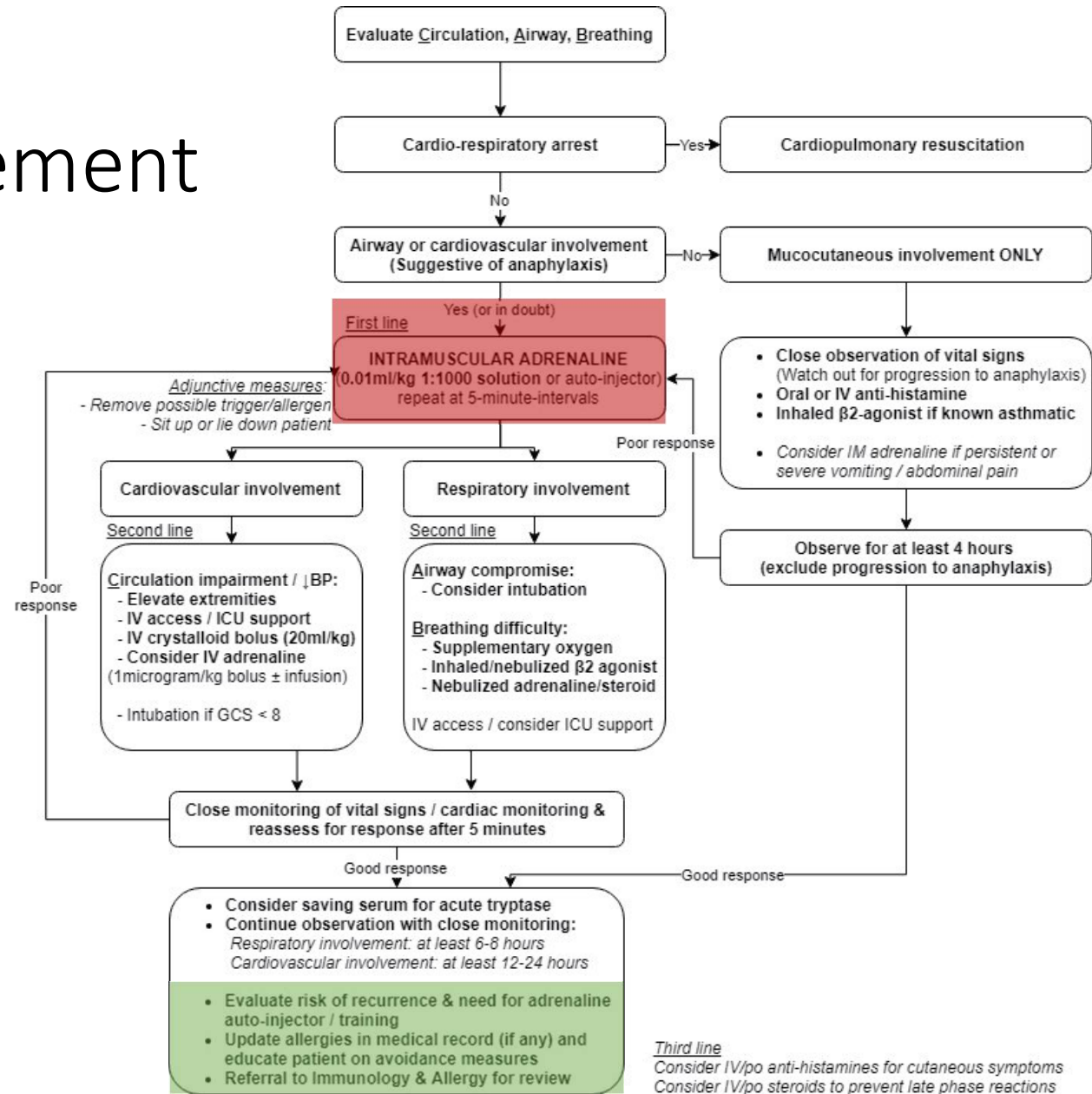
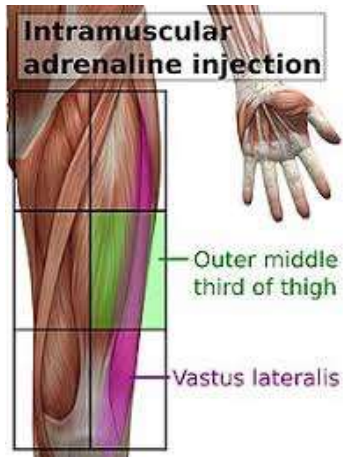


Positive: Either **>11.4ng/mL** OR **(Baseline+20%)+2ng/ml**



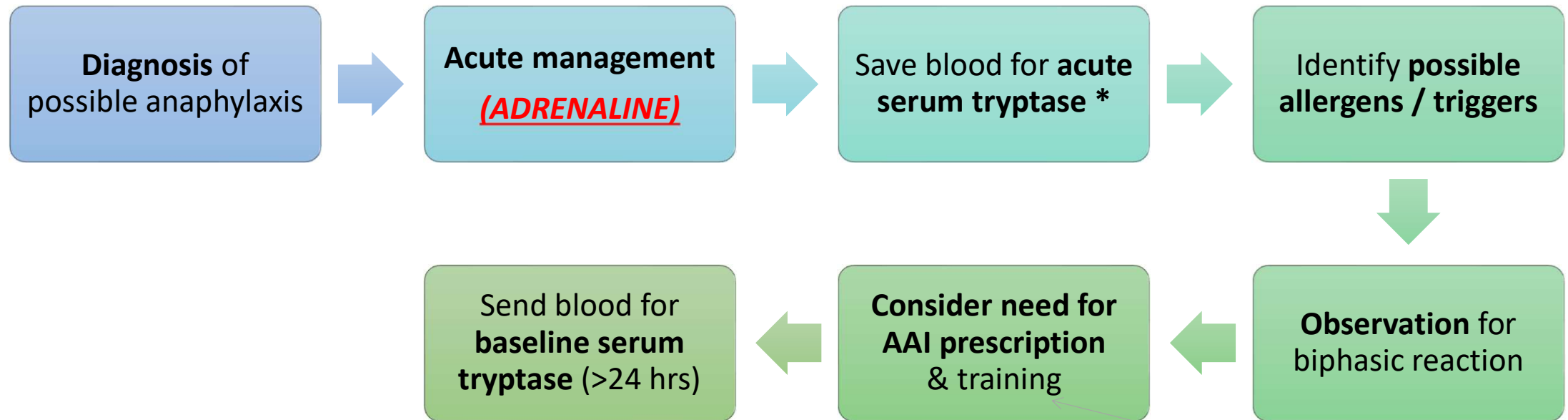
# Histaminergic

## Anaphylaxis: management



# Histaminergic

## Anaphylaxis: management

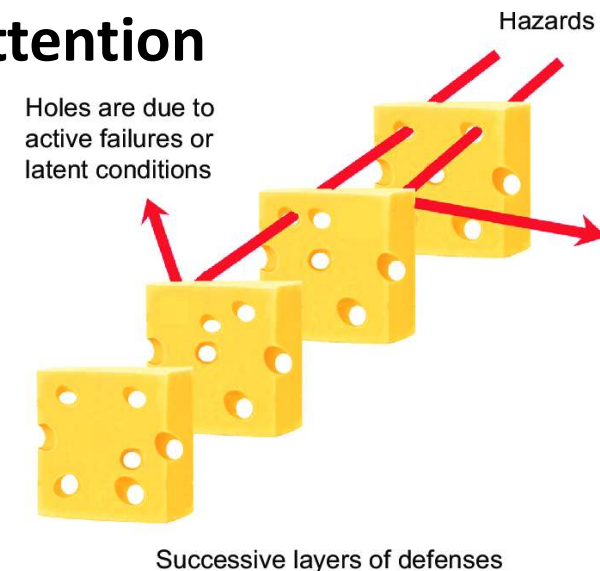


**Respiratory** symptoms or signs should be observed for **at least 6 to 8 hours**  
**Hypotension or collapse** require close monitoring for **12-24 hours**.

# Histaminergic

## Three key points for anaphylaxis survivors

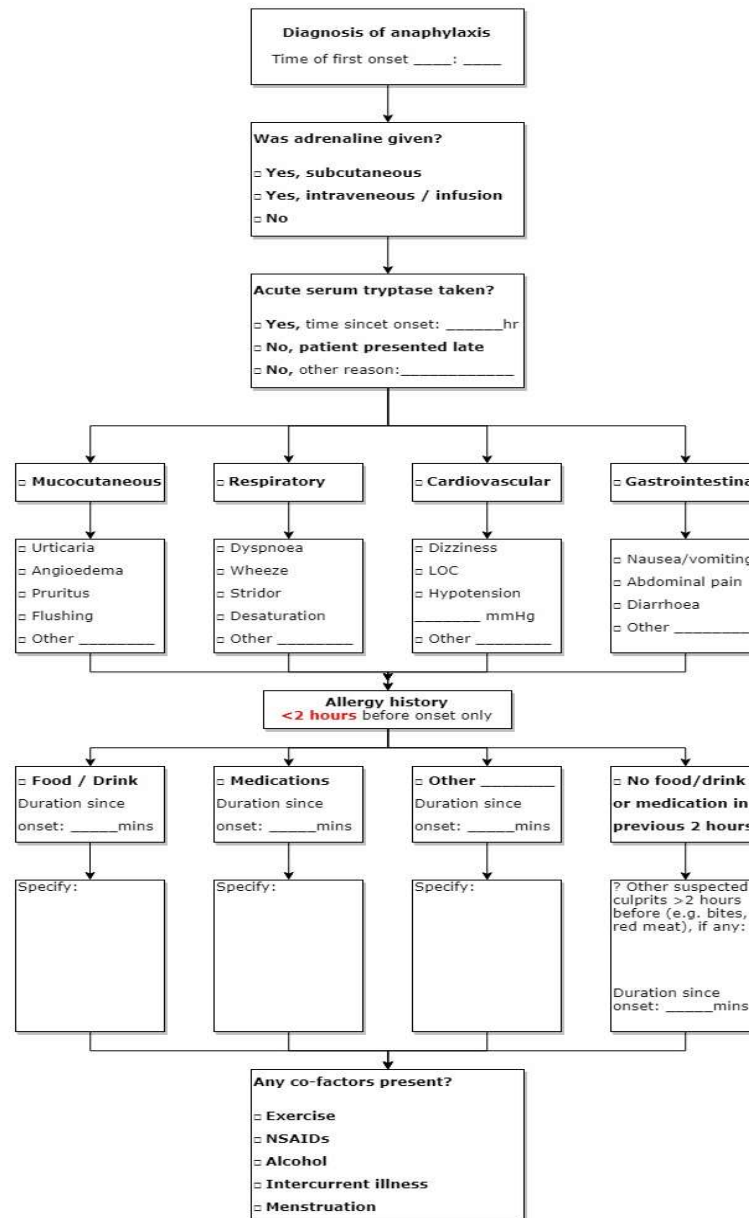
- 1. Strict allergen avoidance**
- 2. Anti-histamines and close observation for mild reactions**
- 3. Adrenaline-autoinjector and emergency medical attention**



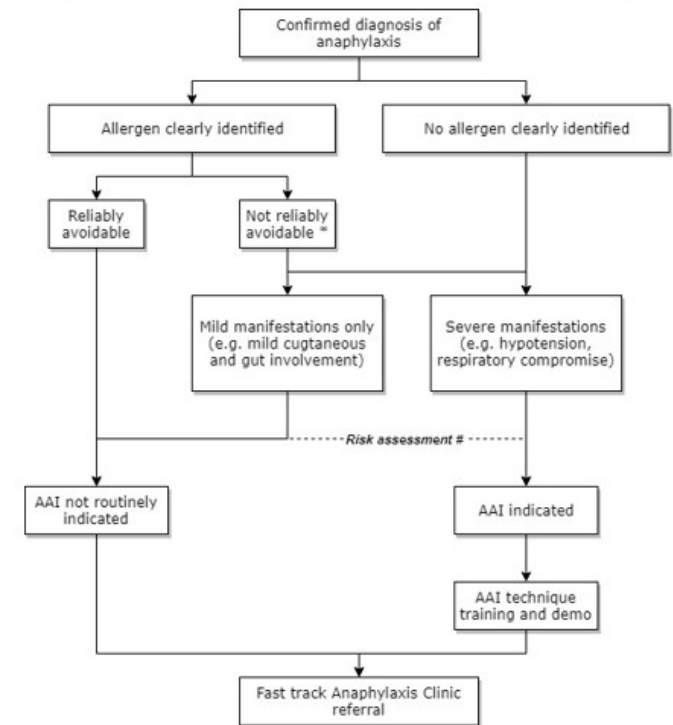
# QMH Anaphylaxis Pathway

## Pre-discharge:

- Comprehensive and targeted **allergy history**
- Ensure **tryptase** taken
- **Patient education**
- **Avoidance advice**
- **Decision for AAI**
- **AAI technique training**
- **Return demonstration**
- **Trigger Allergist referral (QMH Anaphylaxis Clinic)**



## Decision for adrenaline autoinjector (on discharge from AED)



### # Risk assessment, factors to consider including:

- History of previous severe reactions
- Likelihood of re-encountering allergen(s)
- Patient comorbidities
- Patient occupation and exposure (e.g. reaction after trace allergen exposure)
- Social circumstances (e.g. availability of emergency help, patient lifestyle)

\* Examples of reliably avoidable allergens: medications, radiocontrast, shellfish, fruits

Examples of non-reliably avoidable allergens: Ubiquitous foods (egg, milk, wheat, nuts etc.), insect stings

### NB:

- Usually only one autoinjector prescription is sufficient
- Consider switching to alternative agents if patient is on beta-blockade or ACEi
- Inform patient that the decision for AAI is tentative and may be revised according to subsequent testing and allergist recommendation

This child has the following allergies:

Name: \_\_\_\_\_

DOB: \_\_\_\_\_

Photo

**Mild/moderate reaction:**

- Swollen lips, face or eyes
- Itchy/tingling mouth
- Hives or itchy skin rash
- Abdominal pain or vomiting
- Sudden change in behaviour

**Action to take:**

- Stay with the child, call for help if necessary
- Locate adrenaline autoinjector(s)
- Give antihistamine:
- (If vomited, can repeat dose)
- Phone parent/emergency contact

**Watch for signs of ANAPHYLAXIS (life-threatening allergic reaction)**

Anaphylaxis may occur without skin symptoms: ALWAYS consider anaphylaxis in someone with known food allergy who has **SUDDEN BREATHING DIFFICULTY**

**A AIRWAY**

- Persistent cough
- Hoarse voice
- Difficulty swallowing
- Swollen tongue

**B BREATHING**

- Difficult or noisy breathing
- Wheeze or persistent cough

**C CONSCIOUSNESS**

- Persistent dizziness
- Pale or floppy
- Suddenly sleepy
- Collapse/unconscious

**IF ANY ONE (OR MORE) OF THESE SIGNS ABOVE ARE PRESENT:**

- 1 Lie child flat with legs raised (if breathing is difficult, allow child to sit)
- 2 Immediately dial 999 for ambulance and say ANAPHYLAXIS ("ANA-FIL-AX-IS")
- 3 In a school with "spare" back-up adrenaline autoinjectors, **ADMINISTER the SPARE AUTOINJECTOR** if available
- 4 Commence CPR if there are no signs of life
- 5 Stay with child until ambulance arrives, do **NOT** stand child up
- 6 Phone parent/emergency contact

**\*\*\* IF IN DOUBT, GIVE ADRENALINE \*\*\***

You can dial 999 from any phone, even if there is no credit left on a mobile. Medical observation in hospital is recommended after anaphylaxis. For more information about managing anaphylaxis in schools and "spare" back-up adrenaline autoinjectors, visit [sparepensinschools.uk](http://sparepensinschools.uk)

**Emergency contact details:**

1) Name: \_\_\_\_\_

2) Name: \_\_\_\_\_

3) Name: \_\_\_\_\_

4) Name: \_\_\_\_\_

**Parental consent:** I hereby authorise school staff to administer the medicines listed on this plan, including a "spare" back-up adrenaline autoinjector (AAI) if available, in accordance with Department of Health Guidance on the use of AAIs in schools.

Signed: \_\_\_\_\_

Print name: \_\_\_\_\_

Date: \_\_\_\_\_

For more information about managing anaphylaxis in schools and "spare" back-up adrenaline autoinjectors, visit: [sparepensinschools.uk](http://sparepensinschools.uk)

**Additional instructions:**

This BSACI Action Plan for Allergic Reactions is for children and young people with mild food allergies, who need to avoid certain allergens. For children at risk of anaphylaxis and who have been prescribed an adrenaline autoinjector device, there are BSACI Action Plans which include instructions for adrenaline autoinjectors. These can be downloaded at [bsaci.org](http://bsaci.org)

For further information, consult NICE Clinical Guidance CG116 Food allergy in children and young people at [guidance.nice.org.uk/CG116](http://guidance.nice.org.uk/CG116)

This is a medical document that can only be completed by the child's healthcare professional. It must not be altered without their permission. This document provides medical authorisation for schools to administer a "spare" adrenaline autoinjector in the event of the above-named child having anaphylaxis (as permitted by the Human Medicines (Amendment) Regulations 2017). The healthcare professional named below confirms that there are no medical contra-indications to the above-named child being administered an adrenaline autoinjector by school staff in an emergency. **This plan has been prepared by:**

Sign & print name: \_\_\_\_\_

Hospital/Clinic: \_\_\_\_\_

Date: \_\_\_\_\_

瑪麗醫院內科（風濕及臨床免疫科）  
及瑪麗醫院急症科

**過敏性休克緊急措施**



姓名: \_\_\_\_\_

貼紙

身份証號碼: \_\_\_\_\_

緊急聯絡人: \_\_\_\_\_ (電話號碼)

(姓名)

我對以下有過敏: \_\_\_\_\_

我必須遮開: \_\_\_\_\_

同時有哮喘?

☐ 有 (有嚴重反應的風險，需要哮喘藥物伴隨) ☐ 沒有

**輕微的症狀**

- 痕癢
- 皮疹
- 嘴唇或面部腫脹但沒有呼吸困難
- 發癢或流鼻涕/打噴嚏
- 噁心或嘔吐



1. 求助
2. 按規定服用抗過敏藥
3. 觀察任何嚴重症狀

如有疑問  
請致電緊急服務並  
準備腎上腺素 (過敏急救筆)

**嚴重的症狀**

- 失去意識
- 感到頭暈或暈眩
- 呼吸困難
- 喉嚨收縮
- 舌頭腫脹



1. 立即使用腎上腺素 (過敏急救筆)\*
2. 致電緊急服務尋求幫助
3. 如果呼吸困難請坐下，如果感到頭暈或暈眩請躺下
4. 請送往醫院接受治療

\* 如哮喘及呼吸困難，也可使用腎上腺素筆後，再用哮喘藥物

☐ 理解並同意「過敏性休克緊急措施」的內容

☐ 明白「腎上腺素自動注射器 (過敏急救筆)」的使用方法: \_\_\_\_\_

☐ 處方: 抗組織胺藥 \_\_\_\_\_ 及 腎上腺素自動注射器

患者簽署

負責醫護人員簽署

日期 (日/月/年)



# HONG KONG ANAPHYLAXIS CONSORTIUM CONSENSUS STATEMENTS

## ON PRESCRIPTION OF ADRENALINE AUTOINJECTORS

### IN THE ACUTE CARE SETTING

Supported by:



1

Adrenaline autoinjectors should be used as **first-line treatment** and prescribed for all patients at risk of anaphylaxis.

2

If indicated, adrenaline autoinjectors should be **prescribed prior to discharge** from the Accident & Emergency Department and an **allergy referral should be triggered immediately**.

3

The decision for prescribing adrenaline autoinjectors should be based on the severity of previous reactions; including objective signs of respiratory involvement, **objective signs of cardiovascular involvement and multiorgan involvement** (regardless of severity).

4

The decision for prescribing adrenaline autoinjectors should be based on demographics and comorbidities; including **history of asthma or chronic obstructive pulmonary disease**.

5

Patients deemed requiring adrenaline autoinjectors should be offered **avoidance advice** and **prescribed one adrenaline autoinjectors while awaiting allergist review**.

6

After patients are prescribed adrenaline autoinjectors, **demonstration by a healthcare professional or instructional video** and **return demonstration by the patient** are required.

7

The **long-term decision** for the continued need of adrenaline autoinjectors should be **reviewed by an allergist**.

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## Hong Kong Anaphylaxis Consortium Consensus Statements on prescription of adrenaline autoinjectors in the acute care setting

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如果即將過期，需向醫生處方一支新的注射器



[www.allergyhk.org/AAI/](http://www.allergyhk.org/AAI/)

# Histaminergic

## Histaminergic urticaria

Differentials: histaminergic

- “Allergic” type I hypersensitivity reaction; e.g. foods, drugs, venom
  - Mostly IgE-mediated reactions
- Spontaneous or autoimmune urticaria and angioedema (CSU)
  - Anti-FcεR1α antibody (mast cells)
- Inducible urticaria and angioedema

**Table 12.1** Classification of urticaria

| Type                     | Classification                                                                                                                                        |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spontaneous urticarias   | Acute spontaneous urticaria<br>Chronic spontaneous urticaria                                                                                          |
| Inducible urticarias     |                                                                                                                                                       |
| Physical urticarias      | Symptomatic dermographism<br>Delayed pressure urticaria<br>Vibratory urticaria<br>Cold contact urticaria<br>Heat contact urticaria<br>Solar urticaria |
| Other forms of urticaria | Cholinergic urticaria<br>Exercise-induced urticaria<br>Contact urticaria<br>Aquagenic urticaria                                                       |

Urticaria of all types can present with an immediate wheal and flare and/or angioedema, except for delayed pressure urticaria, which is characterized by deep swellings arising with a  $\frac{1}{2}$ –12 h latency and symptomatic dermographism, which does not present with angioedema. (Adapted from Zuberbier T, Asero R, Bindslev-Jensen C, et al. EAACI/GA<sup>2</sup>LEN/EDF/WAO Guideline: Definition, classification and diagnosis of urticaria. Allergy 2009; 64:1417–1426.)

• Urticarias are classified by:

• **Duration**

- Acute (<6 weeks)
- Chronic (≥6 weeks)

• **Clinical presentation**

- Spontaneous
  - No specific trigger or eliciting factor
- Inducible
  - Identifiable trigger(s)

# Histaminergic

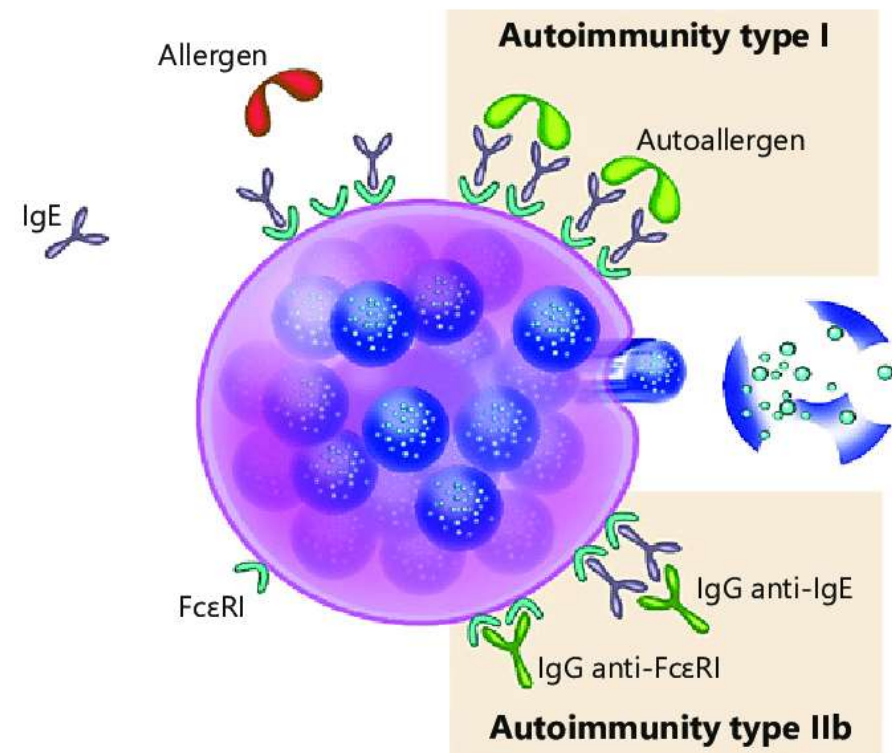
## Chronic Spontaneous Urticaria (CSU)

**'Chronic'** = > 6 weeks

**'Spontaneous'** = i.e. NOT allergy

**'Urticaria'** = hives and/or angioedema

- **No exogenous stimulus** or cause: « Spontaneous »
- Chronic auto-inflammatory skin disease
- Autoantibodies against IgE or FcεRI activate basophils mast cells
  - ➔ **Degranulation: histamine** and other mast cell mediators
  - ➔ Binds to H-receptors located on endothelial cells and sensor nerves
- Concomitant autoimmune thyroid disease in 10%





# Monitoring of disease activity

- Many different tools available, most popular: **Urticaria Activity Score UAS7**
- Number of wheals + Intensity of pruritus over past 7 consecutive days***

| Wheals                                                         | None | 0  |
|----------------------------------------------------------------|------|----|
| Mild (<20 wheals/24 hrs)                                       |      | +1 |
| Moderate (20-50 wheals/24 hrs)                                 |      | +2 |
| Intense (>50 wheals/24 hrs or large confluent areas of wheals) |      | +3 |

| Pruritus                                                                          | None | 0  |
|-----------------------------------------------------------------------------------|------|----|
| Mild (present but not annoying or troublesome)                                    |      | +1 |
| Moderate (troublesome but does not interfere with normal daily activity or sleep) |      | +2 |
| Intense (severe, interferes with normal daily activity or sleep)                  |      | +3 |

**0 points**

A cutoff of 11 points for UAS7 (sum of daily UAS scores over 7 consecutive days) showed sensitivity and specificity of 74% and 86% for change in score, respectively (Hawro 2018)

Copy Results

Next Steps >>>

## 自我記錄及監測病情： 蕁麻疹活性七日量表 (UAS7)<sup>1</sup>

### 1 為蕁麻疹徵狀評分

蕁麻疹活性七日量表 (UAS7) 包括「膨疹數量」和「搔癢程度」兩個參數。患者應按照以下描述，為自己的蕁麻疹徵狀評分。

| 分數     | 膨疹數量                 | 搔癢程度               |
|--------|----------------------|--------------------|
| 0 (沒有) | 沒有膨疹                 | 沒有搔癢               |
| 1 (輕微) | 24 小時內 <20 個         | 有搔癢但不造成困擾          |
| 2 (中度) | 24 小時內 20-50 個       | 搔癢造成困擾，但不影響日常生活或睡眠 |
| 3 (嚴重) | 24 小時內 >50 個或出現大面積膨疹 | 搔癢嚴重，影響到日常生活或睡眠    |

患者會分別得出介乎 0-3 分之間的「膨疹數量」分數和「搔癢程度」分數。

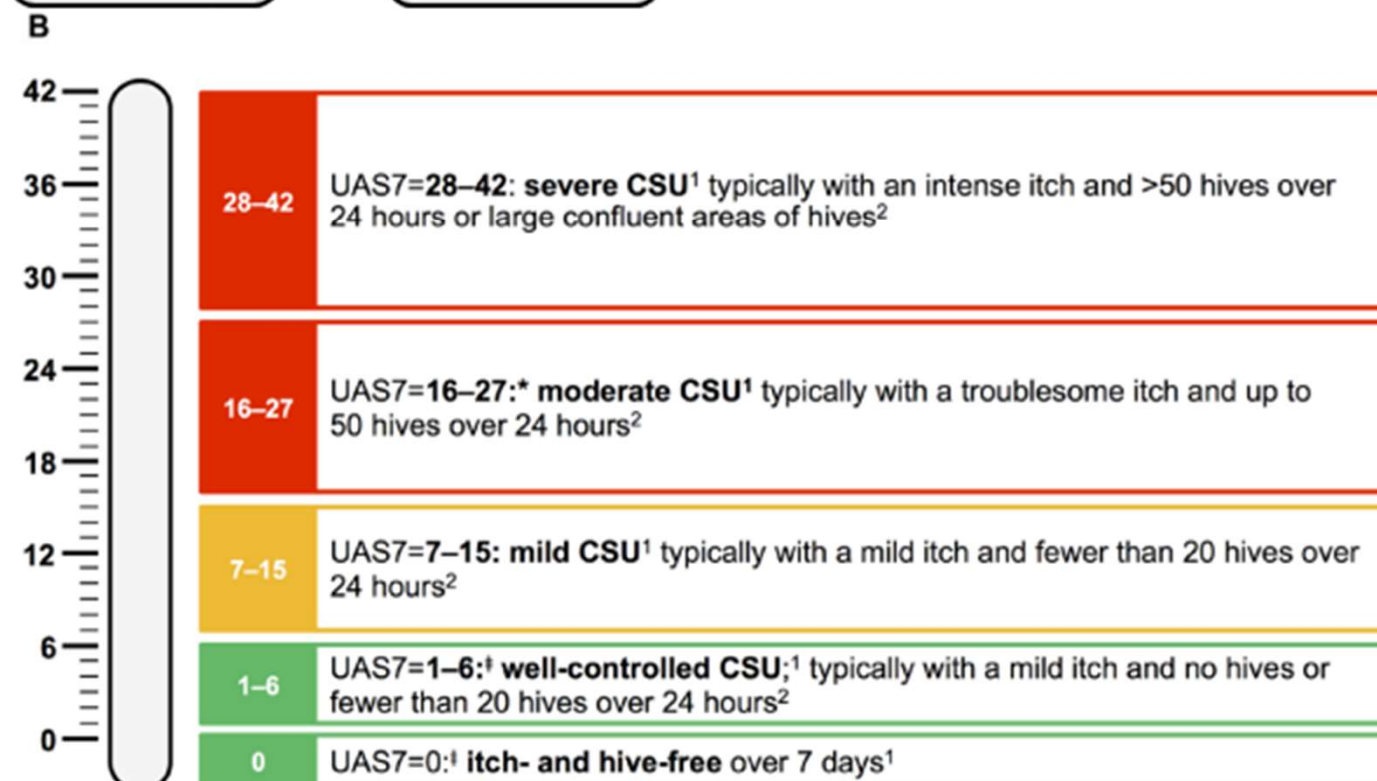
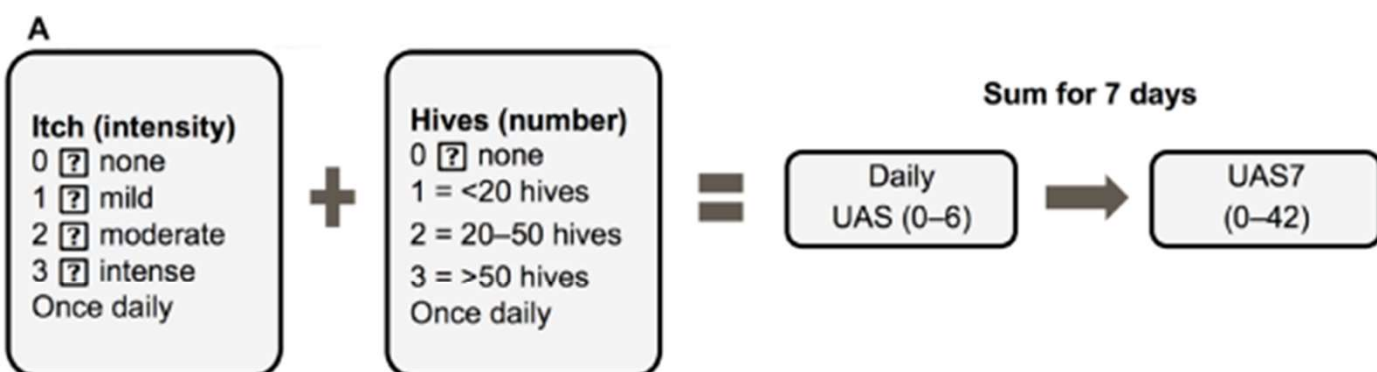
### 2 每日記錄徵狀分數

患者應每日記錄「膨疹數量」分數和「搔癢程度」分數，並將兩者相加，得出 UAS 分數 (0-6 分)，再每 7 日將 UAS 分數相加，得出 UAS7 分數 (0-42 分)。

### 3 定期帶同記錄覆診

完整的 UAS7 分數記錄有助患者持續追蹤自己的蕁麻疹病情。患者亦應定期帶同記錄覆診，方便醫生跟進患者的病情控制及評估治療效果。





## 自我記錄及監測病情： 蕁麻疹活性七日量表 (UAS7)<sup>1</sup>

第  

●

星期

請在適當位置填寫記錄日期，然後根據自己每日的蕁麻疹徵狀，圈出最合適的「**膨疹數量**」和「**搔癢程度**」分數值，並按指示將分數加總。

| 第<br>星期       | 膨疹數量     |                    |                      |                                    | 搔癢程度     |                   |                                    |                                 | UAS 分數<br>(將左邊<br>所圈選的<br>「膨疹數量」<br>分數和<br>「搔癢程度」<br>分數相加) |
|---------------|----------|--------------------|----------------------|------------------------------------|----------|-------------------|------------------------------------|---------------------------------|------------------------------------------------------------|
|               |          |                    |                      |                                    |          |                   |                                    |                                 |                                                            |
|               | 沒有<br>膨疹 | 24<br>小時內<br><20 個 | 24<br>小時內<br>20–50 個 | 24<br>小時內<br>>50 個或<br>出現大面積<br>膨疹 | 沒有<br>搔癢 | 有搔癢<br>但不造成<br>困擾 | 搔癢造成<br>困擾，但<br>不影響<br>日常生活<br>或睡眠 | 搔癢<br>嚴重，影<br>響到日常<br>生活或<br>睡眠 |                                                            |
| 日期            | 沒有       | 輕微                 | 中度                   | 嚴重                                 | 沒有       | 輕微                | 中度                                 | 嚴重                              |                                                            |
| 第 1 日 ( / / ) | 0        | 1                  | 2                    | 3                                  | 0        | 1                 | 2                                  | 3                               |                                                            |
| 第 2 日 ( / / ) | 0        | 1                  | 2                    | 3                                  | 0        | 1                 | 2                                  | 3                               |                                                            |
| 第 3 日 ( / / ) | 0        | 1                  | 2                    | 3                                  | 0        | 1                 | 2                                  | 3                               |                                                            |
| 第 4 日 ( / / ) | 0        | 1                  | 2                    | 3                                  | 0        | 1                 | 2                                  | 3                               |                                                            |
| 第 5 日 ( / / ) | 0        | 1                  | 2                    | 3                                  | 0        | 1                 | 2                                  | 3                               |                                                            |
| 第 6 日 ( / / ) | 0        | 1                  | 2                    | 3                                  | 0        | 1                 | 2                                  | 3                               |                                                            |
| 第 7 日 ( / / ) | 0        | 1                  | 2                    | 3                                  | 0        | 1                 | 2                                  | 3                               |                                                            |

**UAS7 分數**  
 (將以上 7 日的 UAS 分數相加)



# Treatment

Identify and eliminate eliciting factors  
(if any)

Pharmacological treatment

- 4 lines of treatment

Regarding anti-histamines (1st line therapy)

- Second generation anti-histamines only
- To be taken regularly
- **Avoid different anti-histamines** at same time
- Up-dosing **up to fourfold**
- **Safety data** available for use of several years continuously

Accepted: 18 December 2017

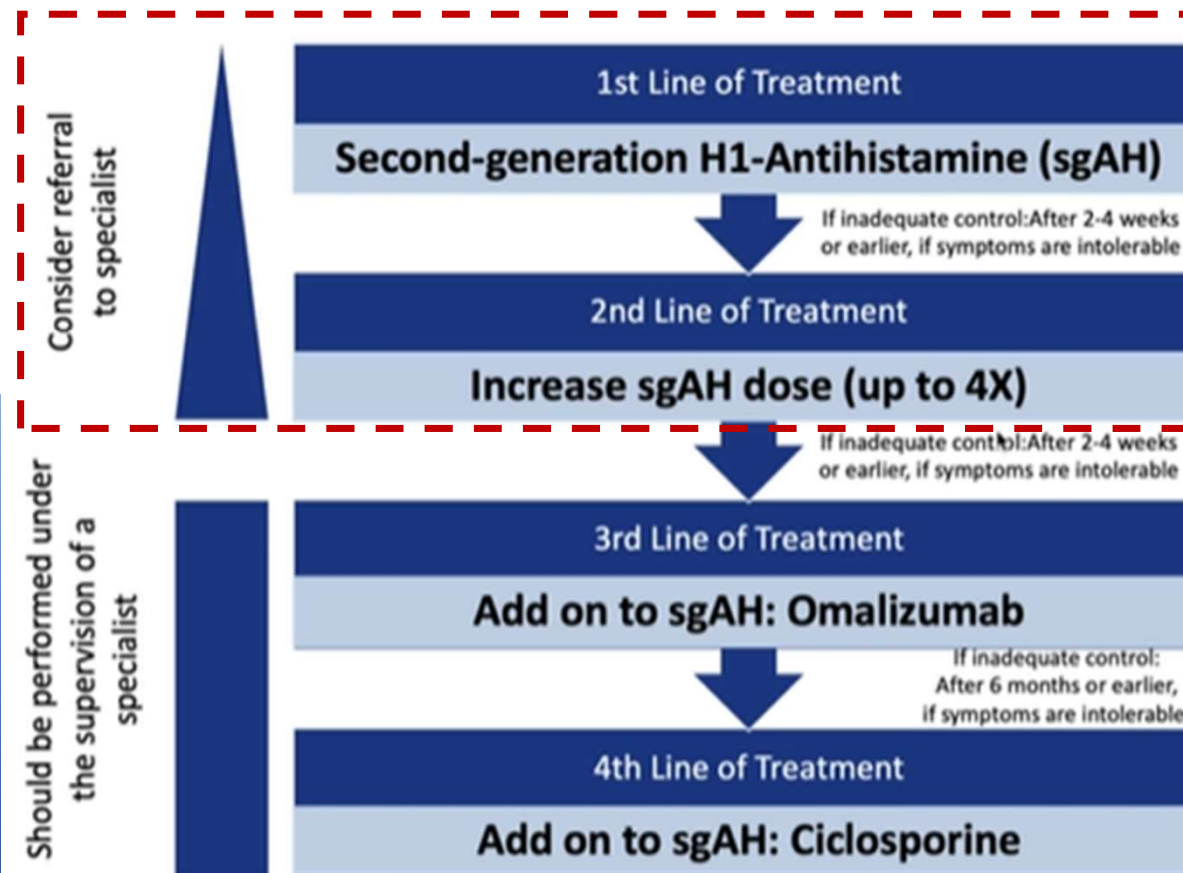
DOI: 10.1111/all.13397



POSITION PAPER

WILEY Allergy

The EAACI/GA<sup>2</sup>LEN/EDF/WAO guideline for the definition, classification, diagnosis and management of urticaria



Adapted from: Zuberbier et al. Allergy. 2018; 73:1393-1

# Treatment

Identify and eliminate eliciting factors  
(if any)

Pharmacological treatment

- 4 lines of treatment

3<sup>rd</sup> line and beyond:  
Personalized therapy

Consider Immunologist  
Referral!

Accepted: 18 December 2017

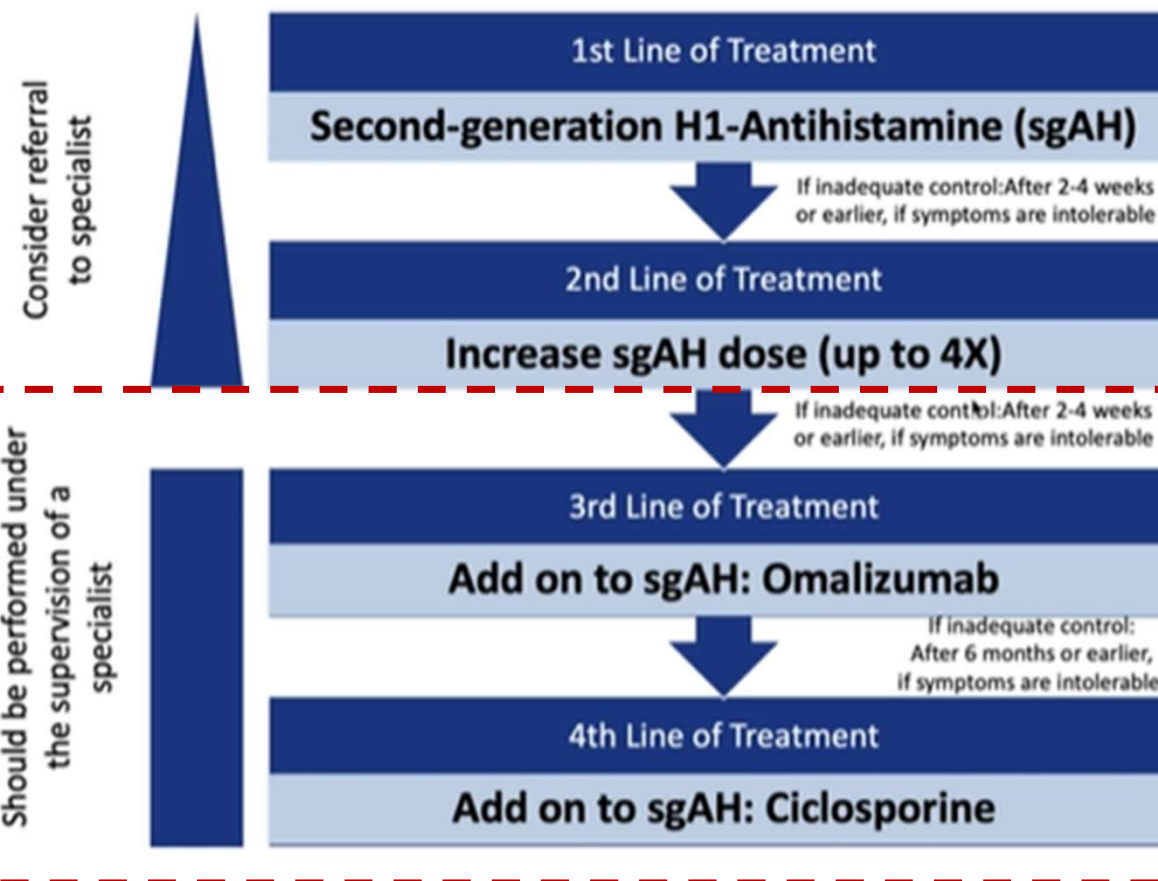
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WILEY Allergy

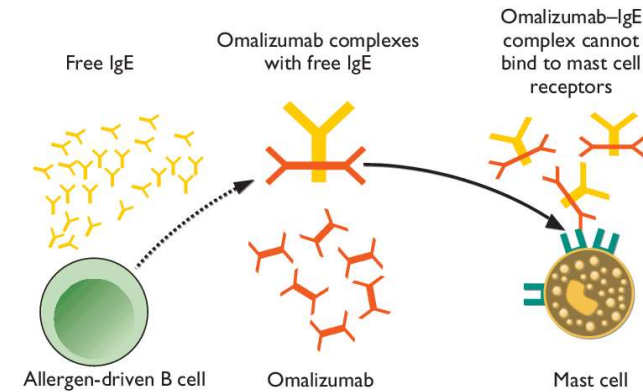
The EAACI/GA<sup>2</sup>LEN/EDF/WAO guideline for the definition, classification, diagnosis and management of urticaria



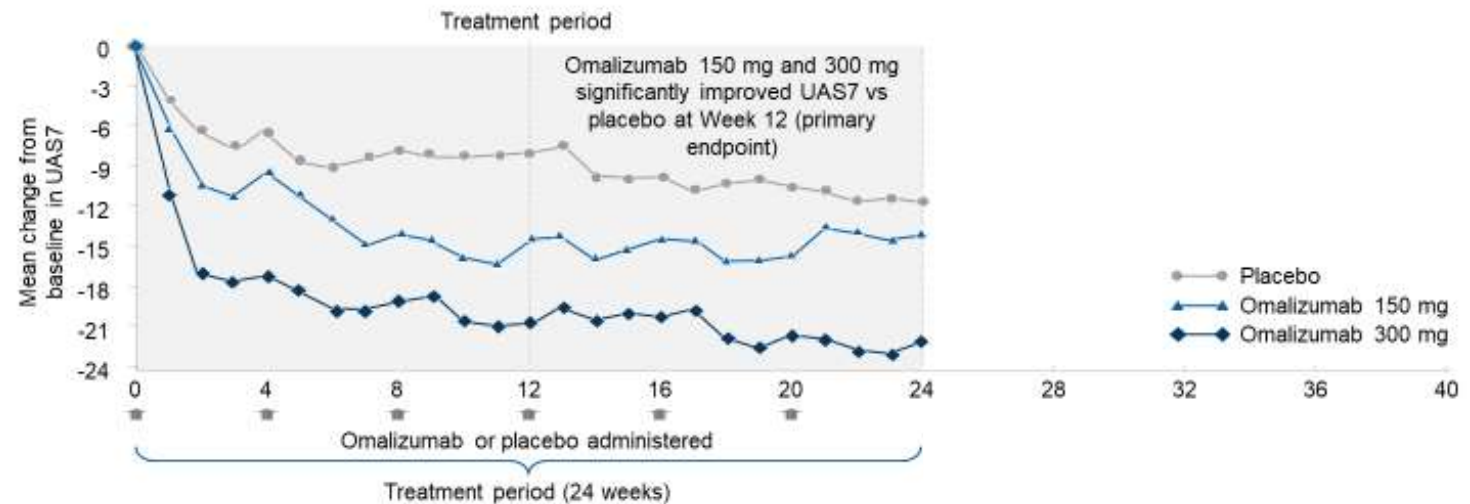
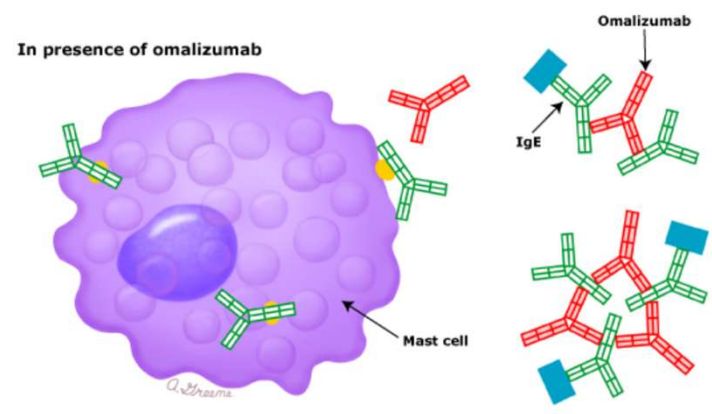
Adapted from: Zuberbier et al. Allergy. 2018; 73:1393-1

# Omalizumab

- Unlike for asthma, a fixed dose approved for CSU
- 300mg every month via subcutaneous injection
- ! *Indications and dosing may vary from region to region*

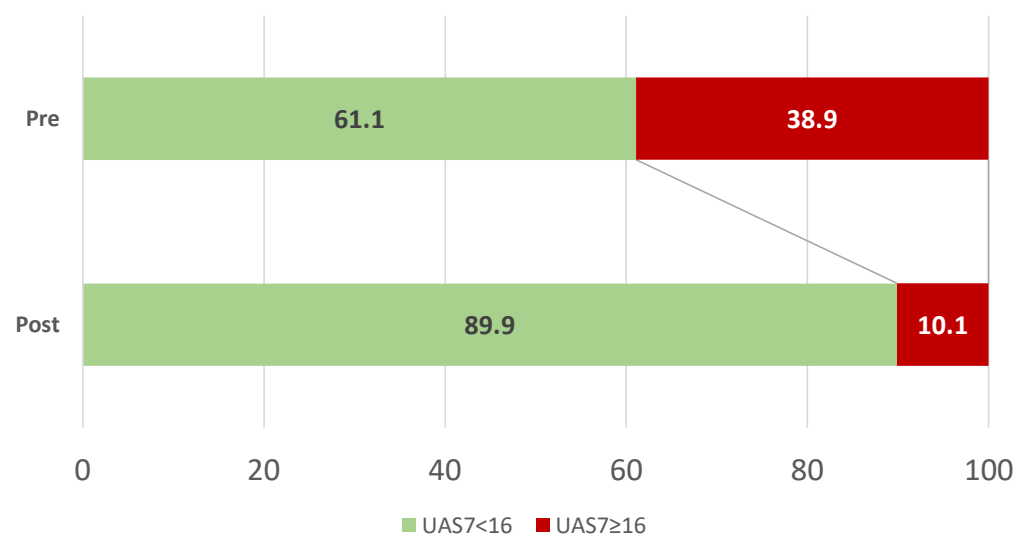


**In *ASTERIA I*, omalizumab resulted in rapid and sustained improvements in UAS7**

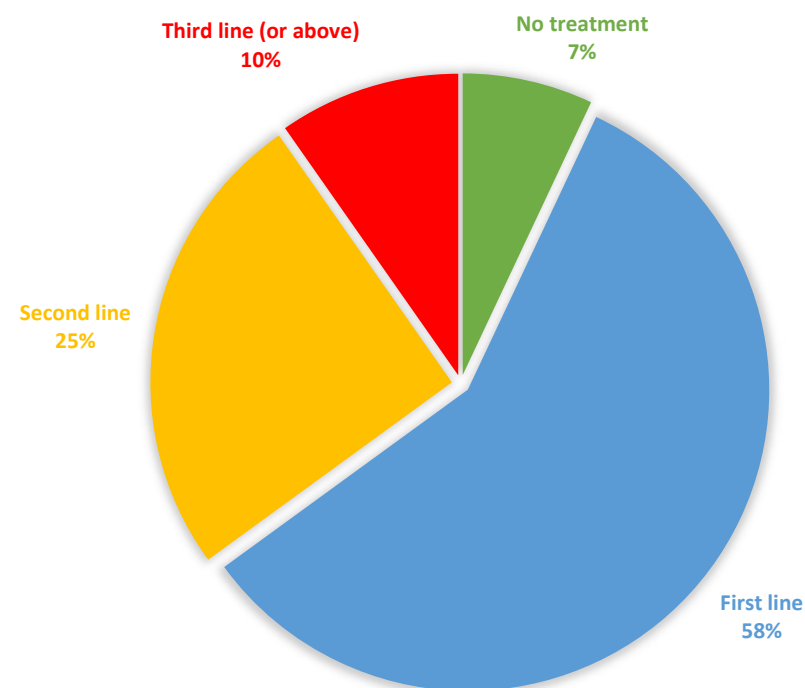


## Role of Immunologist on Chronic Spontaneous Urticaria

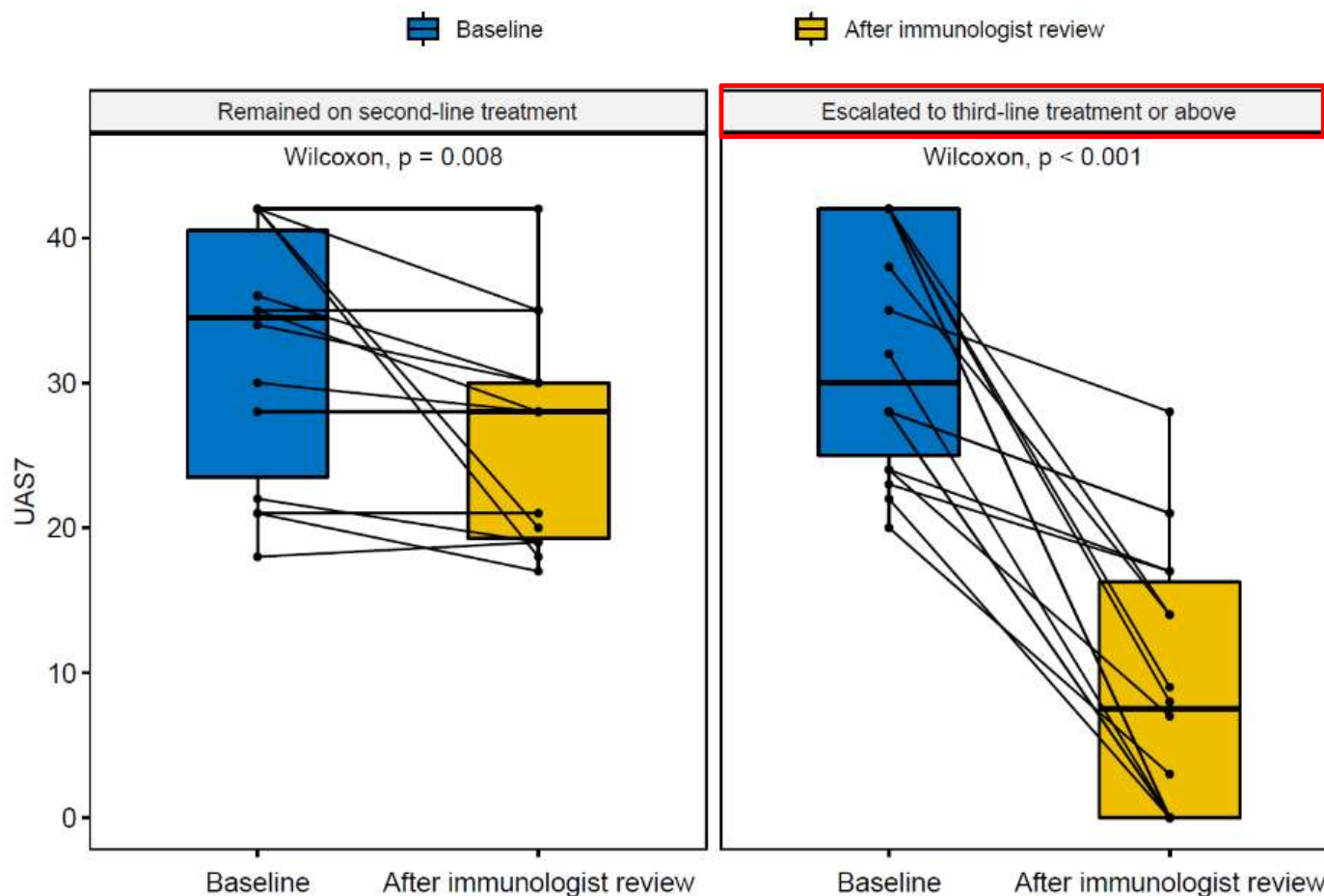
Treatment pre- and post- Immunologist review



Breakdown of treatment modalities in Urticaria Clinic



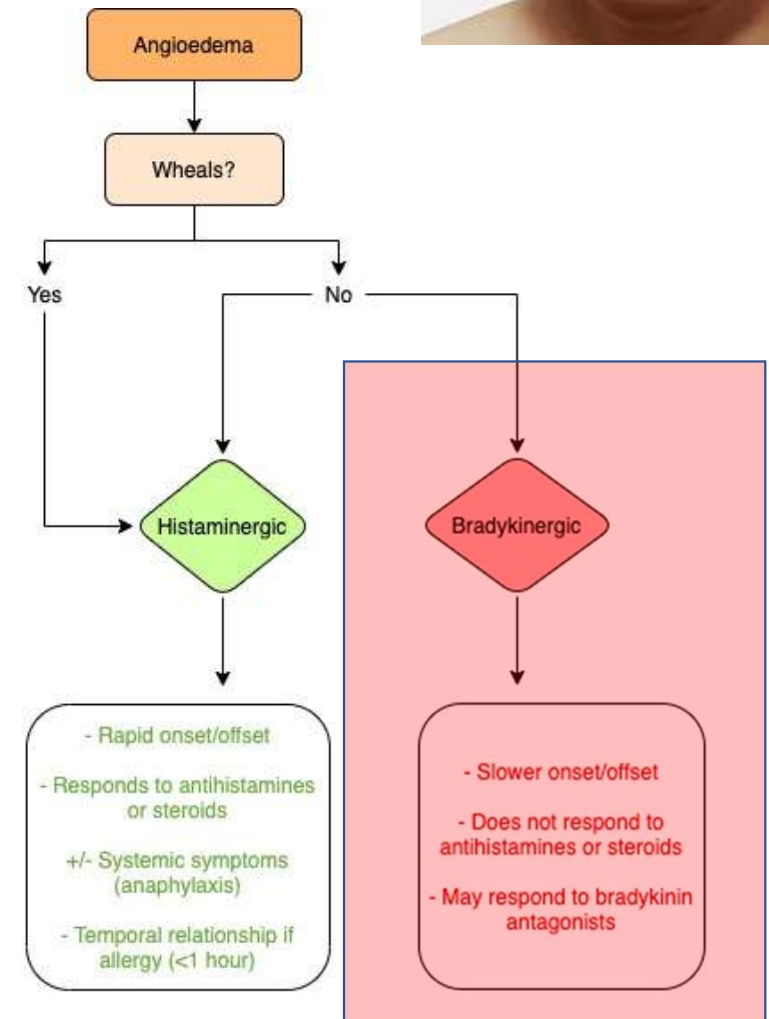
## Role of Immunologist on Chronic Spontaneous Urticaria





# Differentials: bradykinergic

- **ACEi-induced angioedema** to be excluded first
  - ~0.1% of ACEi recipients
- Hereditary angioedema (Type I,II,III ...and more)
  - 1:10,000 population
  - Reduced level or function of **C1-esterase inhibitor**
- Acquired C1-esterase inhibitor deficiency
  - 1:100,000 population
  - Onset >40 years old, associated with malignancy, ↓C1q level



# Take home messages

**“URTICARIA” = WHEELS AND/OR ANGIOEDEMA**



## 1. Histaminergic reactions

- Mast cell mediated
- Clinical presentation ranges from wheals only to anaphylaxis
- **Most commonly: Spontaneous vs. allergic reactions**
- **Anaphylaxis = acute, systemic, life-threatening allergic reaction**
- **Histaminergic urticaria**
  - Chronic spontaneous urticaria
  - Inducible urticaria

## 2. Bradykininergic angioedema

- Angioedema without wheals
- Exclude ACE-inhibitor exposure first



Questions?

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